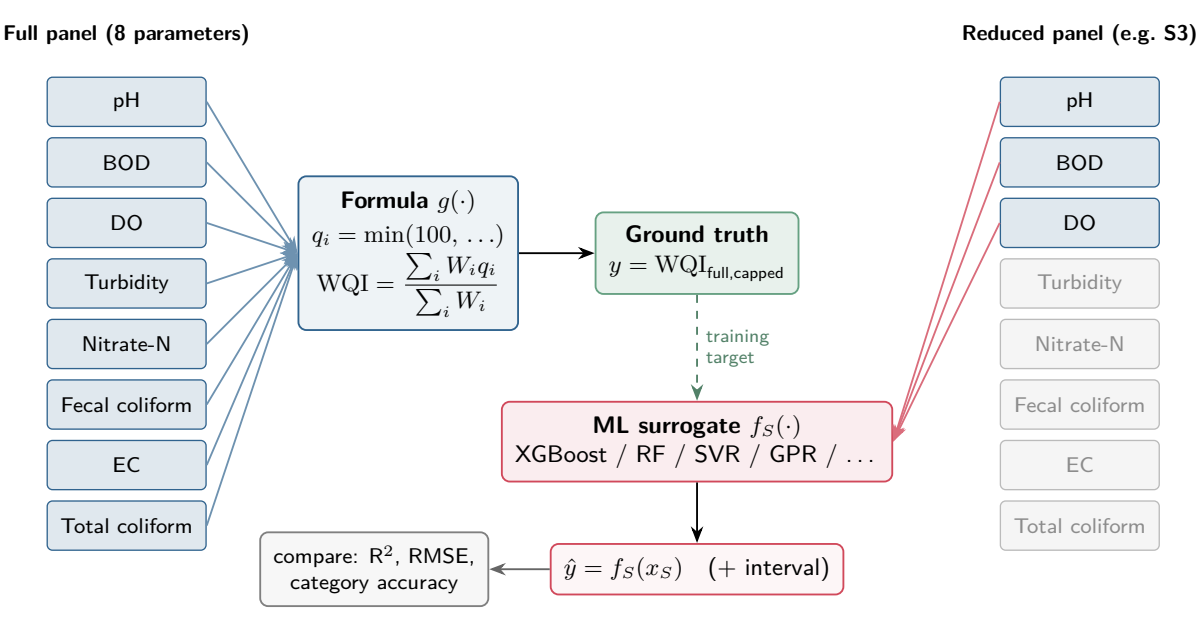
Full panel (8 parameters)
pH
BOD
DO
Turbidity
Nitrate-N
Fecal coliform
EC
Total coliform
Formula $g(\cdot)$
$q_i = \min(100, \ldots)$
$\mathrm{WQI} = \frac{\sum_i W_i q_i}{\sum_i W_i}$
Ground truth
$y = \mathrm{WQI}_{\text{full,capped}}$
training target
ML surrogate $f_S(\cdot)$
XGBoost / RF / SVR / GPR / ...
$\hat{y} = f_S(x_S)$ (+ interval)
compare: R^2, RMSE, category accuracy
Reduced panel (e.g. S3)
pH
BOD
DO
Turbidity
Nitrate-N
Fecal coliform
EC
Total coliform